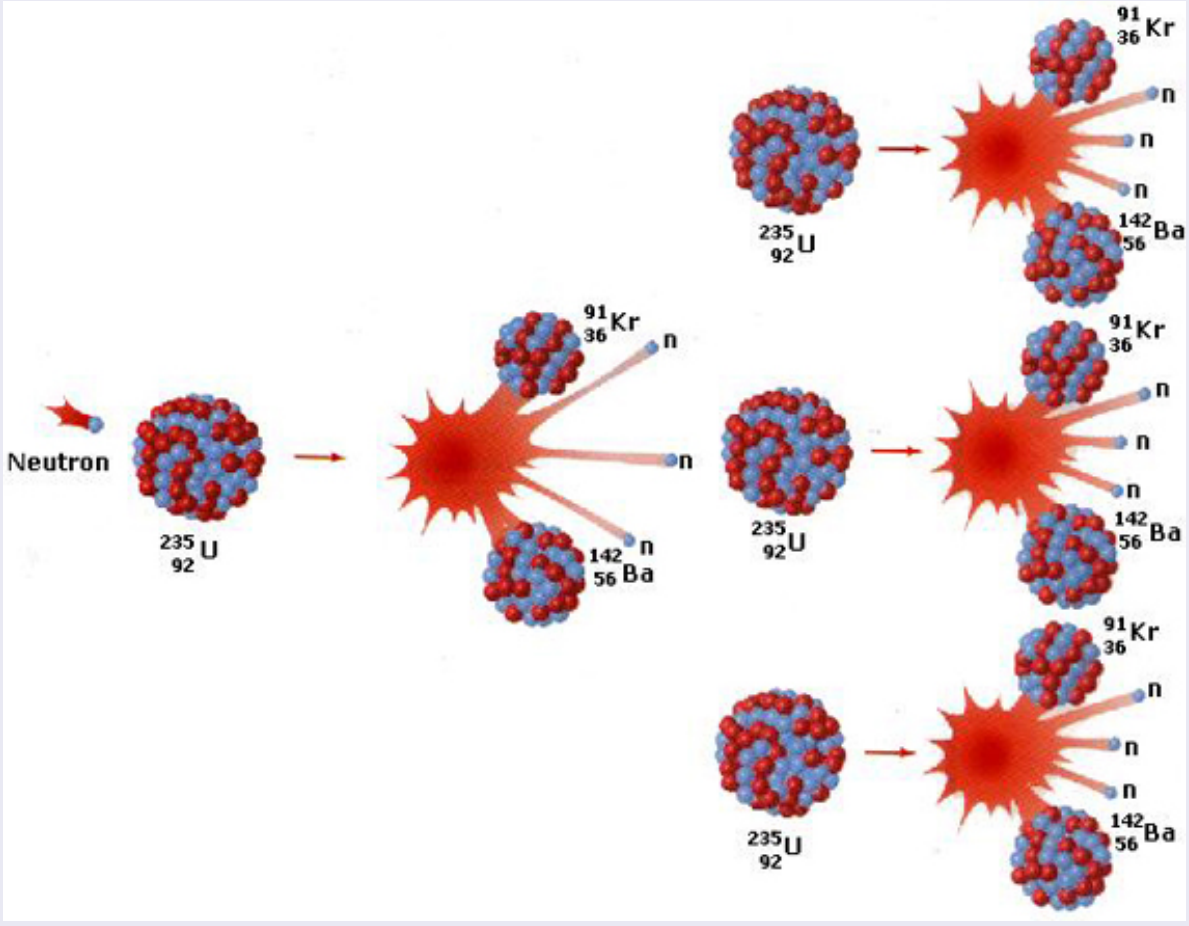
Introduction to Nuclear Engineering and Monte Carlo Simulations

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Fission chain reaction



Elements of Reactor Physics

Effective multiplication factor k_{eff}

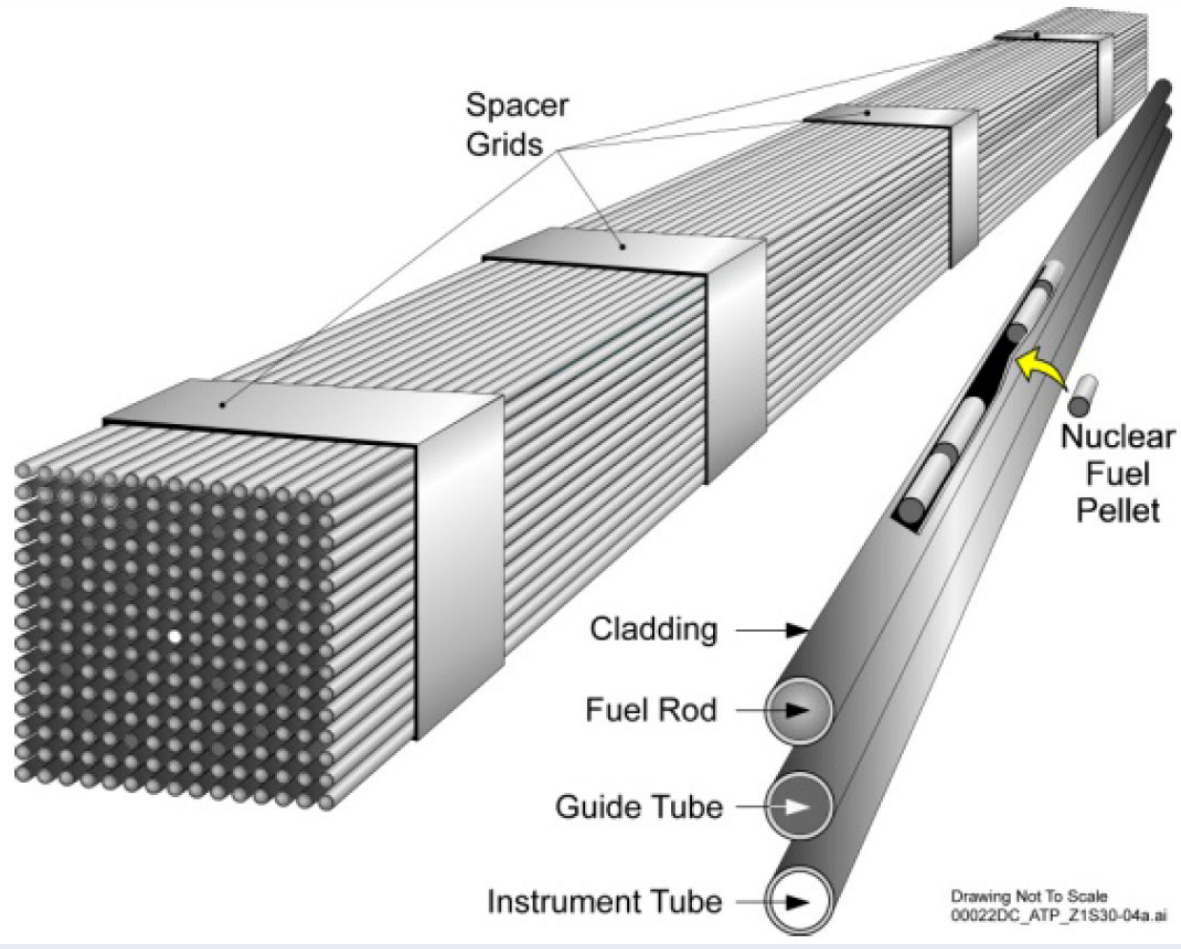
k_{eff} is a factor by which populations of two successive generations differ. It is also called the k-eigenvalue of the system.

Critical, sub-critical, super-critical system

- When $k_{eff} = 1$ then the system is steady-state, so-called critical.
- When $k_{eff} < 1$ then the population in the system decays over the generations, and the system is so-called sub-critical.
- When $k_{eff} > 1$ then the population in the system grows over the generations, and the system is so-called super-critical.

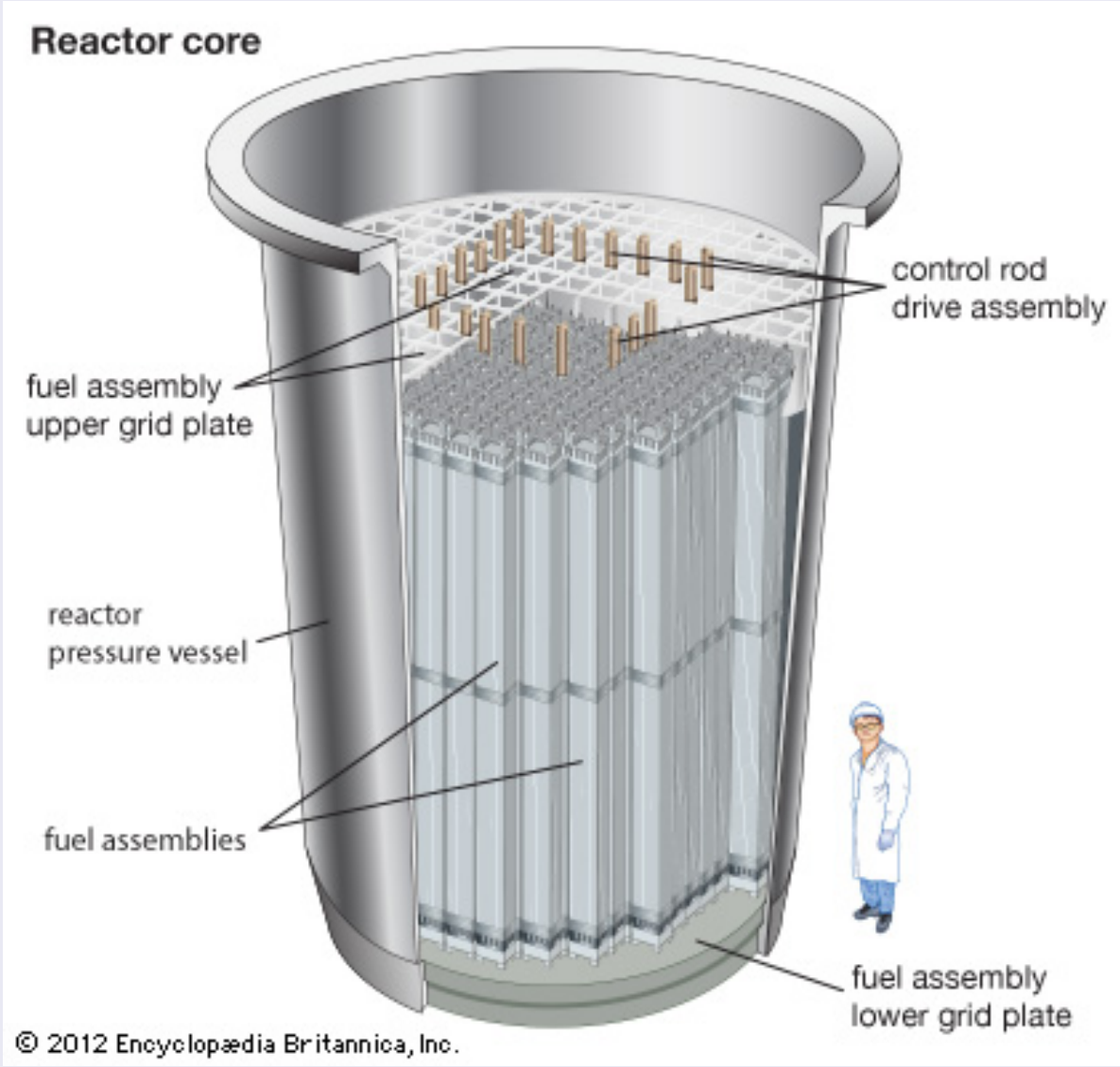
Elements of Reactor Physics

Nuclear fuel assembly



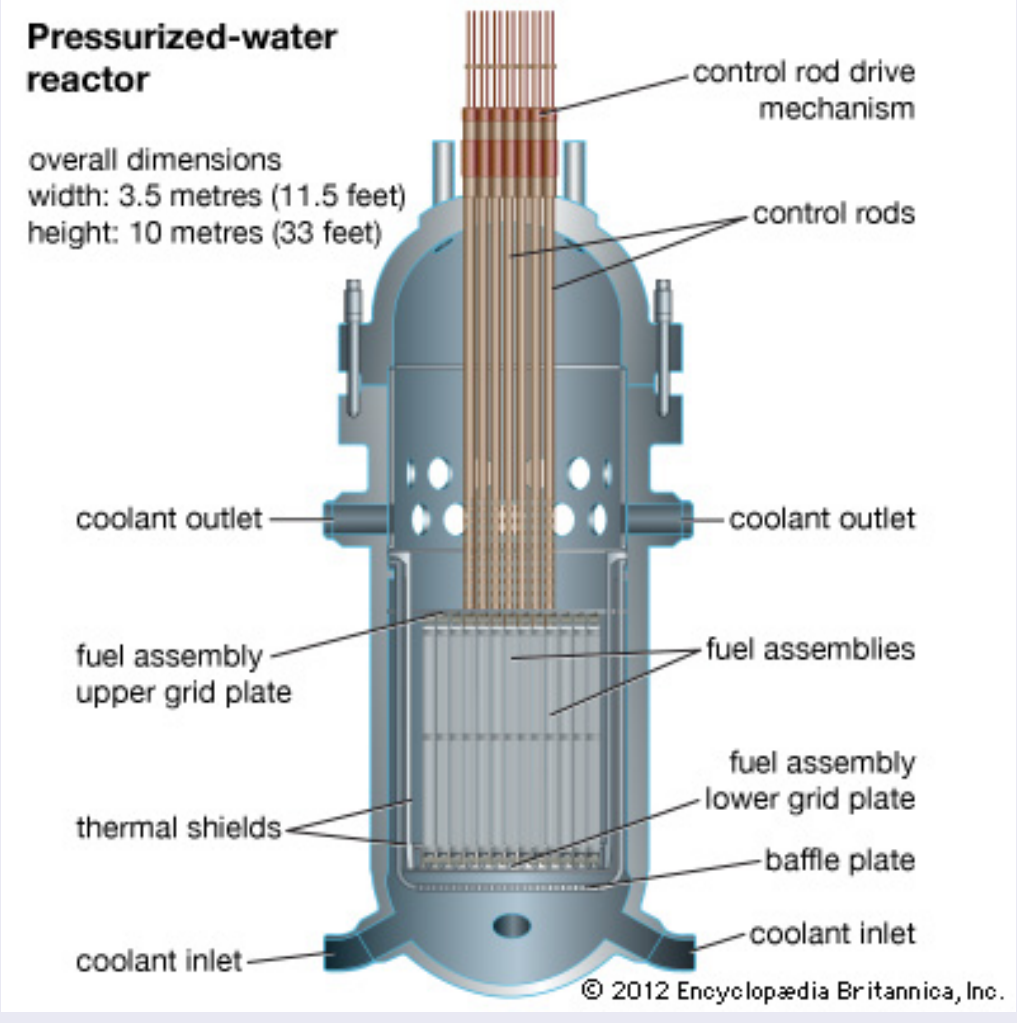
Elements of Reactor Physics

Nuclear reactor core



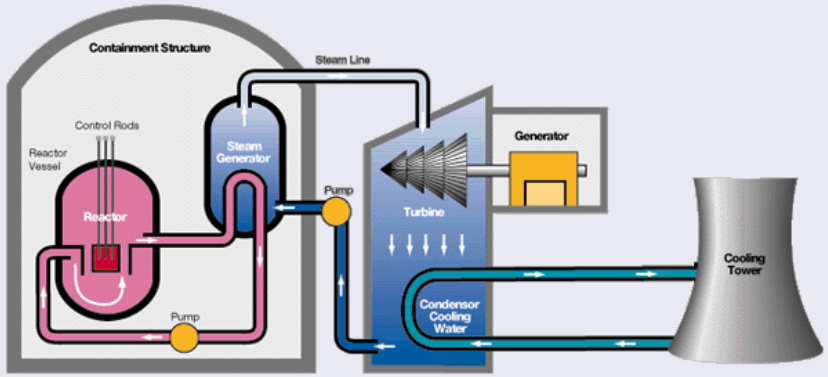
Elements of Reactor Physics

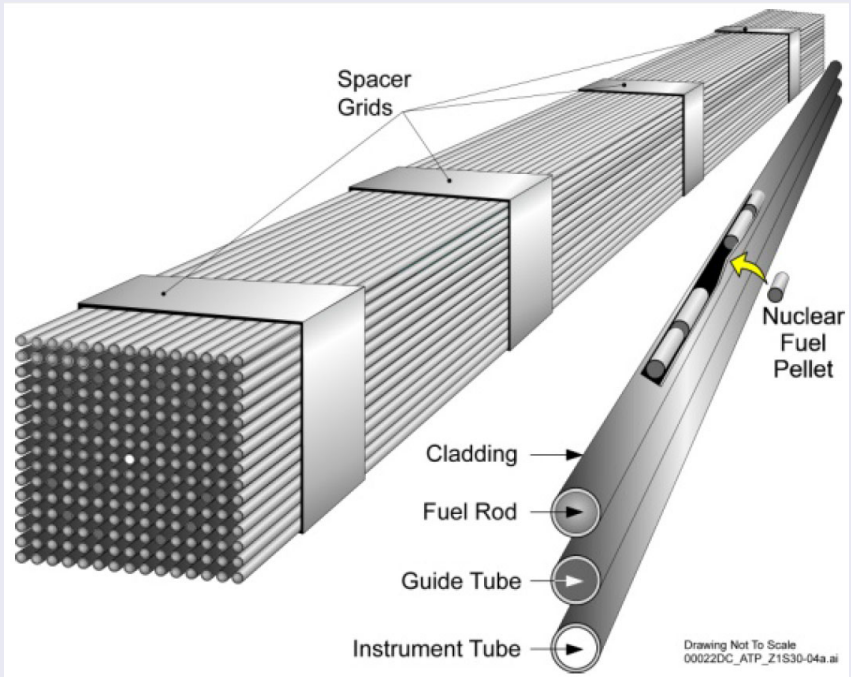
Nuclear reactor



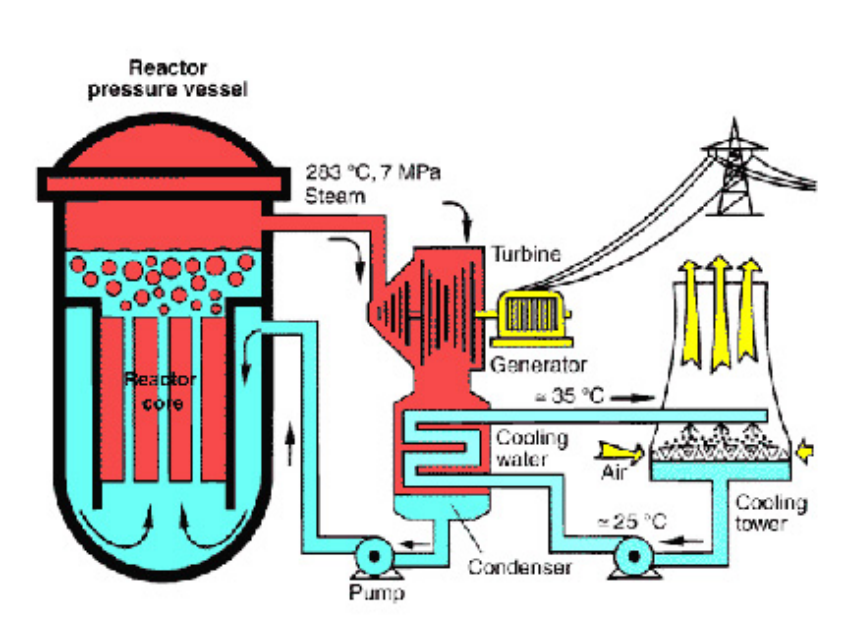
Elements of Reactor Physics

Basic types of nuclear reactors: PWR





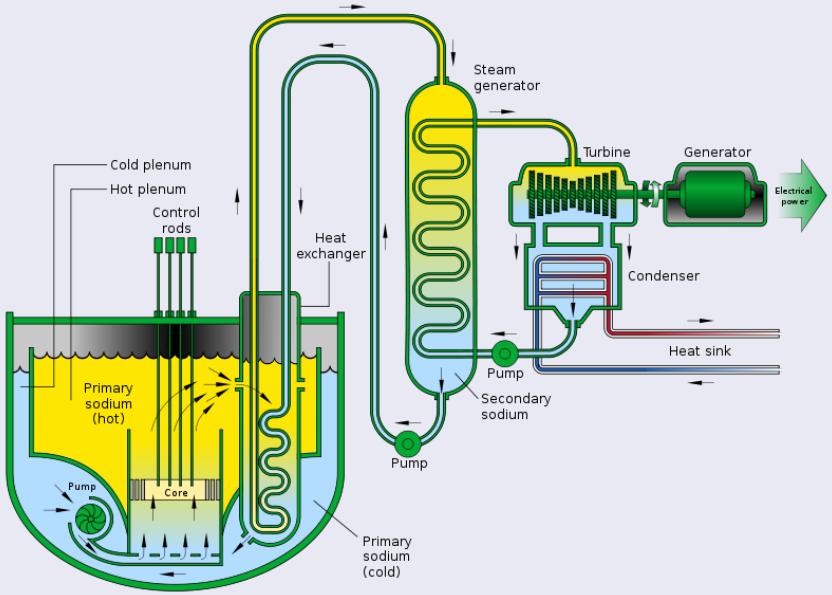
Basic types of nuclear reactors: BWR

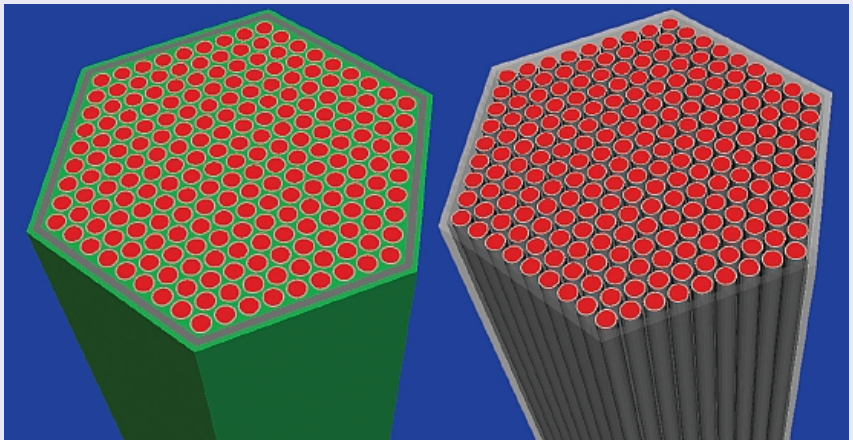




Elements of Reactor Physics

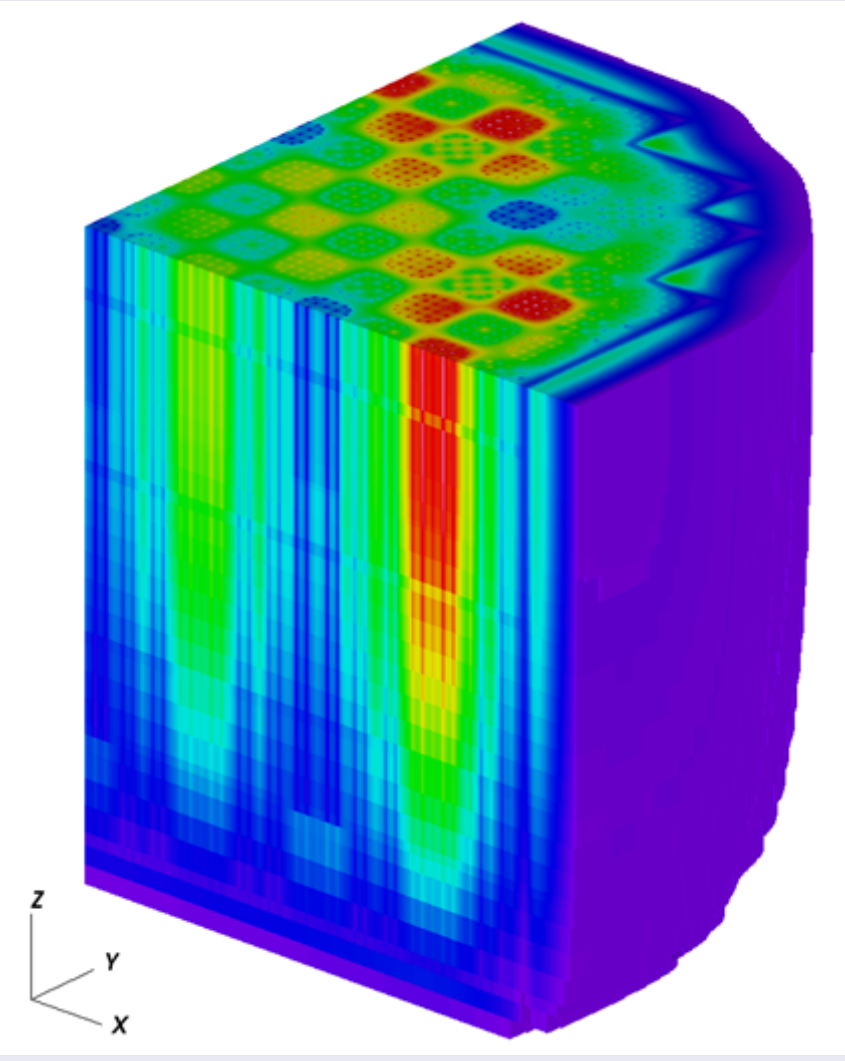
Basic types of nuclear reactors: FR (fast reactors, e.g. sodium cooled)





Elements of Reactor Physics

Power distribution in the reactor core



Elements of Reactor Physics

Neutron flux ϕ

$$\phi(\vec{r}, E, \vec{\Omega}, t) = n(\vec{r}, E, \vec{\Omega}, t)v$$

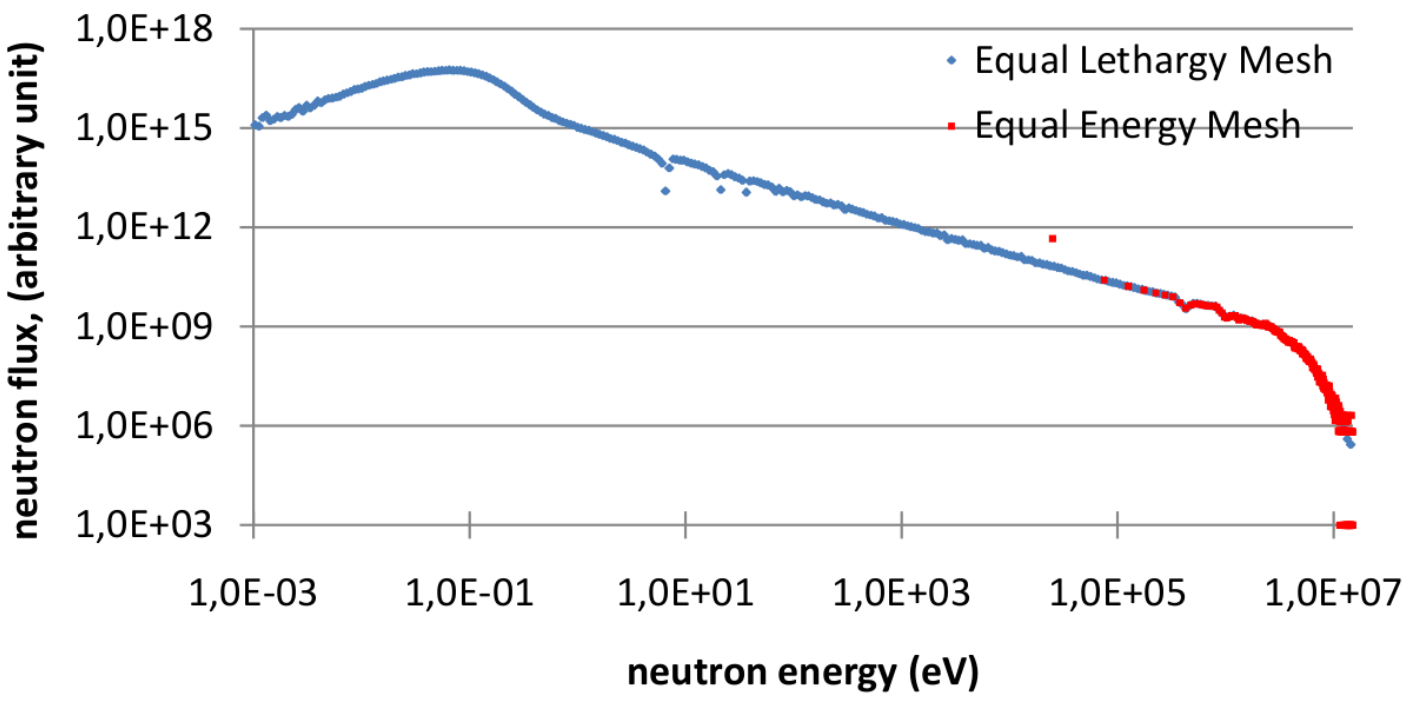
where n is concentration of neutrons at $\vec{r}$ at time t, with energy E and direction $\vec{\Omega}$, and v is the scalar velocity.

Unit: $1/\text{cm}^2\text{s}$

Question

What is the physical meaning of the neutron flux?

Example of neutron flux energy spectrum



Elements of Reactor Physics

Neutron flux energy spectrum

Neutron flux energy spectrum depends on

- type of the reactor (thermal, fast)
- presence of absorbers in the core
- fuel depletion, etc.

What is the Monte Carlo method?

What is the Monte Carlo method?

The Monte Carlo method is a way of solving a deterministic problem by a stochastic approach using random numbers.

A number n of independent observations (e.g. neutron histories) are collected. The result is derived from the averaged observation.

The Monte Carlo method is commonly applied on deterministic problems that are **difficult** to solve by deterministic methods.

Methodological position

		Type of the problem	
		deterministic	stochastic
solution method	deterministic	numerical analysis	probability theory
	stochastic	**Monte Carlo**	simulation

Advantages and disadvantages of the Monte Carlo method

Advantages

- simplicity (e.g. the transport equation does not have to be formulated to find the neutron flux in the reactor)
- applicable on complex problems without simplifications (e.g. accurate geometry, continuous-energy cross sections)
- efficiency in complex problems (the statistical error generally decays as $n^{-1/2}$)

Disadvantages

- computational cost (depending on the complexity of the problem)
- results with statistical errors

Major Monte Carlo codes

Major Monte Carlo codes for reactor physics

- MCNP - Monte Carlo N-Particle Transport Code (Los Alamos National Laboratory, USA)
- TRIPOLI (CEA, France)
- KENO (Oak Ridge National Laboratory, USA)
- SERPENT (VTT, Finland)
- OpenMC (MIT, USA)